

Module 1: Algebra & Functions

Ph.D. Math Camp

Algebra Review

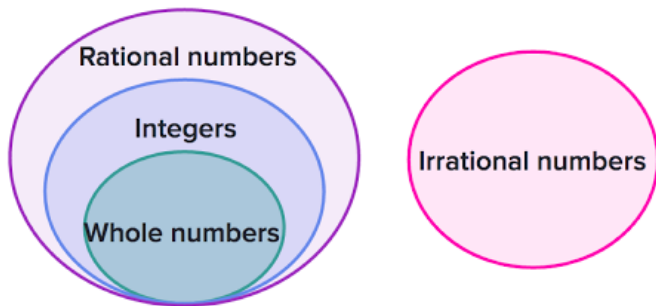
Numbers

- ▶ **Numbers.** We can have **counting numbers** or **natural numbers** ($\mathbb{N}$): 1, 2, 3, 100, 50000, ... or **Whole Numbers** which includes 0 together with the counting numbers.
- ▶ **Integers** ($\mathbb{Z}$): are whole numbers but can be negative (-1,-2,-3,...)
- ▶ **Rational Numbers** ($\mathbb{Q}$): ratio of two integers (i.e., a rational number can be written as a fraction). If you can write a number as a ratio of two integers (e.g., 1 over 10, -5 over 23, 1 over 3, etc.), it is a rational number.
- ▶ **Irrational Numbers.** These would be decimals that do not stop and do not repeat. Some important irrational numbers are π or e .
- ▶ **Real Numbers** ($\mathbb{R}$) are all rational and irrational numbers.
 - ▶ $\mathbb{R}^+$: set of all positive real numbers
 - ▶ $\mathbb{R}^-$: set of negative real numbers

Numbers

- ▶ Numbers
- ▶ Integers
- ▶ Rational Numbers
- ▶ Irrational Numbers
- ▶ Real Numbers

Figure: Types of Numbers

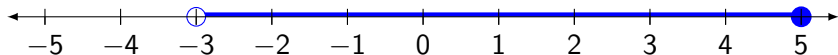


Inequalities and Intervals

- ▶ An inequality is a comparison of unequal terms
- ▶ Real number a is _____ real number b
 - ▶ Greater than $>$
 - ▶ Less than $<$
 - ▶ Greater than or equal to $\geq$
 - ▶ Less than or equal to $\leq$
- ▶ **Intervals** are a set of real numbers that can be represented on a number line by a line segment. Inequalities are often used to describe intervals.
 - ▶ Intervals may be finite or infinite in extent and may or may not contain either endpoint.
 - ▶ Often intervals are written in brackets and parentheses

Inequalities and Intervals

- ▶ To show inclusive of endpoints, use brackets: $[3,5]$
- ▶ Exclusive of the endpoints use parenthesis: $(3,5)$
- ▶ Inclusive of the right endpoint only use the parenthesis on the left and the bracket on the right $(3, 5]$
- ▶ On a number line, an interval inclusive of the endpoints has solid dots on the endpoints, whereas an interval exclusive of the endpoints has open dots on the endpoints. Take $(-3, 5]$ as an example:



Absolute Value

The absolute value of a number can be interpreted geometrically as its **distance from zero on a number line**.

- ▶ Also, the distance between any two numbers is **the absolute value of their difference** (in either order).
- ▶ Absolute value is represented as $| \quad |$

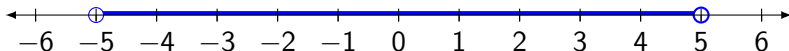
For the interval $[-3,5]$ what is the distance between the endpoints?

$$| -3 - 5 | = | -5 - 3 | = | -8 | = 8$$

Combining Absolute Value and Inequalities

Combine the concepts of absolute value and inequalities and solve for x in some basic expressions involving two operations.

1. Find the set of solutions for $|x| < 5$. In English: **We are looking for real numbers x whose distance from the origin (zero) is less than 5 units.** This is the interval $(-5, 5)$.



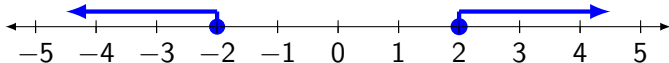
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2. Solutions to $|x| \geq 2$. In English: which numbers, x , are at least 2 units **away** from the origin?



Combining Absolute Value and Inequalities

3. Find the solution to $|x - 2| \leq 1$

$$-1 \leq x - 2 \leq 1.$$

Solve for x : $2 - 1 \leq x \leq 1 + 2 \rightarrow \boxed{1 \leq x \leq 3}$

4. Find the solution to $|2x - 5| < 2$

$$-2 < 2x - 5 < 2. \text{ Solve:}$$

$$-2 + 5 < 2x < 2 + 5 \implies \boxed{\frac{3}{2} < x < \frac{7}{2}}$$

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Exponential Notation

The following rules define the expression a^n for $a > 0$ and all rational values of n :

- ▶ Integer Powers: If n is a positive integer, then $a^n = a \cdot a \cdot a \cdots a$, where this product contains n factors.
- ▶ Fractional Powers: If n and m are positive integers, then $a^{\frac{n}{m}} = (\sqrt[m]{a^n}) = (\sqrt[m]{a})^n$
- ▶ Negative powers: $a^{-n} = \frac{1}{a^n}$
- ▶ Zero power: $a^0 = 1$

Remember these as you will use them repeatedly.

Some examples:

1. $3^0 = 1$
2. $27^{-2/3} = (27^{\frac{1}{3}})^{-2} = (3)^{-2} = \frac{1}{9}$

Laws of Exponents

If a is a real number

- ▶ The product law: $a^r \cdot a^s = a^{r+s}$
- ▶ The quotient law: $\frac{a^r}{a^s} = a^{r-s}$
- ▶ The power law: $(a^r)^s = a^{rs}$

Some examples:

1. $\frac{1}{r^5} \cdot \frac{1}{r^5} = r^{-10}$
2. $\beta^7 \cdot \beta^{-3} = \beta^4$

Factoring

- ▶ Distributive law: For any real numbers a , b , and c :
 $ab + ac = a(b + c)$.
The key to factoring is spotting common terms.

Factoring

- ▶ Polynomials: You will most often be factoring polynomials, expressions of the form $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$.
- ▶ Here x is a variable, n is a nonnegative integer, and $a_0, a_1, a_2 \dots a_n$ are real numbers or coefficients of the polynomial.
- ▶ If $a_n \neq 0$, n is said to be the **degree** of the polynomial. For example, $8c^7 + 6c^4 + c^2 + 8$ is a polynomial of degree 7.
 - ▶ *Constants* do not vary in an expression or across individuals or units of observations. *Variables* can take on different values. For instance, we may have a coefficient on a variable we are trying to solve for that can take on different values. Variables can take on different values within a given set.

Factoring

3. To factor an expression is to write it as a product of two or more simpler terms, called **factors**.
- ▶ Factoring a natural number means expressing it as a product of smaller natural numbers.
 - ▶ Factoring a polynomial means expressing it as a product of polynomials of a lower degree.
 - ▶ Factoring is used to simplify complicated expressions and solve equations and is based on the distributive law of multiplication over addition.

Factoring

- ▶ We will most likely be looking at polynomials of degree 2 (also known as quadratics).
- ▶ The goal of factoring a quadratic is to go from an expression of the form $a_2x^2 + a_1x + a_0$ to one of the form $(cx + a)(dx + b)$, with a, b, c, d being integers.
- ▶ Clearly, c and d are factors of a_2 . Also, a and b are factors of a_0 .
 - ▶ As a short-cut that sometimes works, you want to find two numbers that, when multiplied, equal to $a_2 \times a_0$ and, when added, equal to a_1 .

Factoring

Try these

1. $x^2 + 11x + 24$.

2. $p^2 - 22p + 40$.

Factoring

1. $x^2 + 11x + 24$. In this case, the two numbers are 8 and 3. Set up the next step of the solution as $(x + 8)(x + 3)$.
2. $p^2 - 22p + 40$. The two numbers are -2 and -20 $\implies (p - 2)(p - 20)$.

Factoring

There is also a special kind of polynomial that **is the difference of two perfect squares**

- ▶ For any real numbers a and b , $a^2 - b^2 = (a + b)(a - b)$.

Notice how the ab terms cancel when you multiply the RHS.

Another key way that you can use factoring is in simplifying fractions by factoring both the numerator and the denominator and divide each by common terms.

Solving Equations via Factoring

Some Examples: Simplify the following expressions

1.
$$\frac{7(q+4)^2(q-1)^4 - 14(q+4)^3(q-1)^2}{(q+4)(q-1)^2}$$

2.
$$\frac{3(x-2)^2(x+1)^2 - 2(x-2)(x+1)^3}{(x-2)^4}$$

3. $x^2 + x - 6 = 0$

4. $y^2 - 10y + 16 = 0$

5. $(s^2 - 81) = 0$

Solving Equations via Factoring

$$1. \frac{7(q+4)^2(q-1)^4 - 14(q+4)^3(q-1)^2}{(q+4)(q-1)^2} \\ = 7(q+4)[(q-1)^2 - 2(q+4)]$$

Factor a $(q+4)$ and $(q-1)^2$ from top and bottom.

$$2. \frac{3(x-2)^2(x+1)^2 - 2(x-2)(x+1)^3}{(x-2)^4} = \frac{(x+1)^2(x-8)}{(x-2)^3}$$

$$3. x^2 + x - 6 = 0 \quad (x+3)(x-2) = 0 \implies (x+3) = 0 \text{ and/or} \\ (x-2) = 0$$

$$4. y^2 - 10y + 16 = 0 \implies (y-8) = 0 \text{ and/or } (y-2) = 0$$

$$5. (s^2 - 81) = 0 \implies (s+9)(s-9) = 0$$

Solving Equations via Factoring

- ▶ Formally, a **quadratic equation** takes the form $ax^2 + bx + c = 0$ ($a \neq 0$).
- ▶ A quadratic equation can be factored into 2 linear expressions. Thus, we say that this type of equation has at most two real solutions or 'roots.'
- ▶ If we are unable to factor the equation with integer coefficients to find the solution, then we can always use the quadratic formula to solve quadratic equations.

Find the solution or roots with the following formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The term $D = b^2 - 4ac$ is called the **discriminant** of the quadratic equation. The sign of D informs us on the number of solutions and the type of solutions.

- ▶ $D > 0$ means the equation has two distinct real solutions
- ▶ $D = 0$ means the equation has one distinct real solution (a real double root)– bounces off or just touches x-axis
- ▶ $D < 0$ means the equation has two distinct complex conjugate solutions.

Example:

1. $3x^2 - 41x + 26 = 0$

$$3x^2 - 41x + 26 = 0$$

$$x = \frac{41 \pm \sqrt{41^2 - 4(3)(26)}}{2(3)} = \frac{41 \pm \sqrt{1681 - 312}}{6}$$

$$x = \frac{41 \pm \sqrt{1369}}{6} = \frac{41 \pm 37}{6}$$

$$\Rightarrow x = \frac{41+37}{6}; x = \frac{41-37}{6}$$

$$\Rightarrow x = \frac{78}{6}; x = \frac{4}{6}$$

$$\Rightarrow \boxed{x = 13} \quad \boxed{x = \frac{2}{3}}$$

System of Equations

- ▶ A group of equations to be solved is called a system of equations.
 - ▶ Geometrically, each equation represents a surface in some space.
 - ▶ The coordinates of any point on the surface solve that equation. A point solves the system of equations if it is on all surfaces.
 - ▶ Therefore, the common intersection of all surfaces contains all the solutions to the system of equations.
- ▶ Linear equations (those representing straight lines) are much less complicated than non-linear equations. And the simplest linear system is one with two equations and two variables.
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System of Equations

There are two ways to solve systems like these:

1. Elimination by multiplication/addition ;
2. Elimination by substitution.

Example: Find real numbers p and q that satisfy the system

$$(1) \quad 8p - 3q = 7$$

$$(2) \quad -p + 7q = 19$$

System of Equations

Let's start with **Elimination by multiplication/addition**:

$$(1) \quad 8p - 3q = 7$$

$$(2) \quad -p + 7q = 19$$

1. Let's decide to eliminate p by multiplying equation (2) by 8.
Note that by multiplying both sides of the equality we aren't changing the solution.

$$(2a) \implies -8p + 56q = 152$$

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$$\implies \boxed{q = 3}$$

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 $\implies \boxed{q = 3}$

3. Plug $q = 3$ into either (1) or (2) to get
 $-p + 7(3) = 19 \implies -p = -2 \implies \boxed{p = 2}$

System of Equations

Now let's solve the same system using the second method:

Elimination by substitution.

1. Solve either (1) or (2) for either p or q . Let's use (2) to solve for p .

$$-p + 7q = 19 \implies -p = 19 - 7q \implies p = 7q - 19$$

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2. Plug the new expression for p into (1) and solve for q .

$$8(7q - 19) - 3q = 7 \implies 56q - 152 - 3q = 7 \implies 53q = 159 \implies \boxed{q = 3}$$

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4. Notice that we get identical answers regardless of which method we use.

System of Equations

Solve for x and y

$$(1) \quad 2x + y = 5$$

$$(2) \quad -x + 4y = -7$$

Changing Parameters in Systems of Two Equations and Two Unknowns

Let's consider two lines in the plane represented by the following two linear equations

$$a_1x + b_1y + c_1 = 0 \quad (1)$$

$$a_2x + b_2y + c_2 = 0 \quad (2)$$

The form of this equation of a line is called the **canonical form**. You may be more familiar with the **slope/intercept form**. It is not difficult to switch between the two and graph them.

$$a_1x + b_1y + c_1 = 0 \quad (1)$$

$$a_2x + b_2y + c_2 = 0 \quad (2)$$

For example, equation (1) can be transformed to slope/intercept form by subtracting a_1x and c_1 from both sides of the equation and dividing by b_1 . The result is

$$y = \left(\frac{-a_1}{b_1}\right)x + \left(\frac{-c_1}{b_1}\right) \quad (3)$$

For the second equation,

$$y = \left(\frac{-a_2}{b_2}\right)x + \left(\frac{-c_2}{b_2}\right) \quad (4)$$

In equation (3) the slope of the line is $m = \frac{-a_1}{b_1}$ and the y-intercept (the vertical coordinate of the point of intersection of the line and the y-axis) is $\frac{-c_1}{b_1}$

Given two lines, there are three possible situations that can occur:

Case 1 : The lines are identical. This happens when the equations are identical (when corresponding coefficients are the same multiple of each other): $a_2 = ka_1$, $b_2 = kb_1$; and $c_2 = kc_1$.
Let's consider

$$4x + 2y - 8 = 0 \quad (5)$$

$$6x + 3y - 12 = 0. \quad (6)$$

Notice that the coefficients of (6) are 1.5 times the coefficients of equation (5).

$$6x + 3y - 12 = 0 \implies 3y = -6x + 12 \implies y = -2x + 4$$

$$4x + 2y - 8 = 0 \implies 2y = -4x + 8 \implies y = -2x + 4$$

$$4x + 2y - 8 = 0 \quad (5)$$

$$6x + 3y - 12 = 0. \quad (6)$$

Case 2: The lines are parallel. This happens when the slopes are identical (when $\frac{-a_1}{b_1} = \frac{-a_2}{b_2} = k$ but y-intercepts differ $\frac{c_1}{b_1} \neq \frac{c_2}{b_2}$)

- If we decrease the constant c_1 by 1 unit (so -9 instead of -8), in equation (5), the new first equation in slope/intercept form is: $y = -2x + \frac{9}{2}$.

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- ▶ If we decrease the constant c_1 by 1 unit (so -9 instead of -8), in equation (5), the new first equation in slope/intercept form is: $y = -2x + \frac{9}{2}$.
- ▶ This is the equation of a line parallel to the line described in equation (6) $y = -2x + 4$.

Case 3: The lines intersect.

However, if we were to change equation (5) by increasing b_1 by 1 and b_2 by 1, the new equations in slope/intercept form are

$$y = \frac{-4}{3}x + \frac{8}{3} \quad \text{and} \quad y = \frac{-3}{2}x + 3$$

If we were to solve this system of equations, we would get a solution of $(x, y) = (2, 0)$.

Example: Try another one.

$$(1) \ 2x + 4y - 8 = 0$$

$$(2) \ x + 2y - 6 = 0$$

Summation Notation

The Greek letter $\sum$ (sigma) is used to express summation.

1. Suppose there is a function $y = x$ but let's say $y = f(x) \implies f(x) = x$. Here $f(1)=1$, $f(2)=2$, etc.
2. Let's now replace x with $i \implies f(i) = i$
3. If we wanted to sum the values of the function $f(i) = i$ from $f(1)$ through $f(n)$

$$\sum_{i=1}^n f(i) = f(1) + f(2) + f(3) + \dots + f(n)$$

$$\sum_{i=1}^n f(i) = \sum_{i=1}^n i = 1 + 2 + 3 + 4 + 5 \dots n$$

Summation Notation

$$\sum_{i=1}^n f(i) = \sum_{i=1}^n i = 1 + 2 + 3 + 4 + 5 \dots n$$

- ▶ $\sum_{i=1}^n f(i)$ is read “Sum of the values of the function $f(i)$ from $f(1)$ through $f(n)$ ”
- ▶ Where i is called the index and ranges between 1 and n .

$$\sum_{k=1}^4 (3k - 1) = [3(1) - 1] + [3(2) - 1] + [3(3) - 1] + [3(4) - 1] = 26$$

Rules for Summation

Rule 1: The summation of a constant k times a variable is equal to the constant times the summation of that variable.

$$\sum_{i=1}^n kX_i = k \sum_{i=1}^n X_i$$

- ▶ Suppose we expand the summation=
 $kX_1 + kX_2 + kX_3 + \dots + kX_n$.
- ▶ Note that we can factor out k and end up with
 $k(X_1 + X_2 + \dots + X_n)$.
- ▶ But $\sum_{i=1}^n X_i = (X_1 + X_2 + \dots + X_n)$.

Rules for Summation

Rule 2: The summation of the sum of observations of two variables is equal to the sum of their summations.

$$\sum_{i=1}^n (X_i + Y_i) = \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i$$

- If we expand the summations $(X_1 + Y_1) + (X_2 + Y_2) + \dots (X_n + Y_n)$. We can rearrange the terms and end up with

$$\sum_{i=1}^n (X_i + Y_i) = \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i$$

Rule 3: The summation of a constant over n observations equals the product of the constant and n

$$\sum_{i=1}^n r = rn$$

Let $n = 4 \implies r + r + r + r = 4r$

- ▶ Another symbol that you may run across is the product notation symbol, $\prod$. This represents a product of function values. For example

$$\prod_{i=1}^3 i^2 = (1^2)(2^2)(3^2) = 1 \cdot 4 \cdot 9 = 36$$

Factorials

- ▶ The **factorial** function $n!$ is defined for every positive integer n as $n! = n(n-1) \cdots 2 \cdot 1$.
- ▶ So, for example, $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$
- ▶ The factorial $n!$ gives the number of ways in which n distinct objects can be arranged in order. Each arrangement is called a permutation.
- ▶ For example, $3! = 6$ since there are six possible permutations of 1, 2, and 3: (1, 2, 3) ; (1, 3, 2) ; (2, 1, 3) ; (2, 3, 1); (3, 1, 2); and (3, 2, 1)
- ▶ Note $0! = 1$

Function Basics

Functions

- ▶ A function is a rule that assigns to each object in a set A exactly one object in a set B .
- ▶ It is often convenient to represent a functional relationship by an equation $y = f(x)$, which we have done. In this context x and y are called variables. Since the numerical value of y is determined from that of x , we refer to y as the **dependent variable** and x as the **independent variable**.
- ▶ Suppose you want to find $f(3)$ if
$$f(x) = 8x - 5 \implies f(3) = 8(3) - 5 = 19$$

Functions

- ▶ Often a function is described as a *mapping* from the domain to the range.

Functions

- ▶ Often a function is described as a *mapping* from the domain to the range.
- ▶ **Functions assign outputs to inputs.** The **domain** of a function is the set of all possible **inputs** for the function, say x . The **range** of a function is the set of its possible **output** values, say y .

Domain and Range

- ▶ For example: What is the domain and range of $y = f(x) = x^2$?
 - ▶ Well, x can take on any value, meaning the domain would be all real numbers.
 - ▶ How about the range? For any x that is mapped into y , you get a nonnegative value. Thus, the range would be $y \geq 0$ (all non-negative real numbers)

Composite Functions

- ▶ There are many situations in which a quantity is given as a function of one variable that, in turn, can be written as a function of a second variable. By combining the functions in the natural way, you can express the original quantity as a function of the second variable. The result is a **composite function**.

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- ▶ **Definition of a composite function:** Given two functions f and g , the **composition** of f and g , denoted $f \circ g$ has the value $f(g(x))$. In other words, we evaluate g at x to obtain $g(x)$ and then evaluate f at $g(x)$ to obtain $f(g(x))$.

Composite Functions

- ▶ Note that $f(g(x))$ is only meaningful if x is in the domain of g and $g(x)$ is in the domain of f .
- ▶ Note that in general, $g(f(x)) \neq f(g(x))$

Example: Given the functions $f(x) = x^2 + 6$ and $g(x) = 2x - 1$, find $(f \circ g)(x)$.

$$\begin{aligned} f(g(x)) &= (2x - 1)^2 + 6 \\ &= (2x - 1)(2x - 1) + 6 = 4x^2 - 4x + 7 \end{aligned}$$

Composite Functions

Example: Environmental Policy

An environmental study of a certain community suggests that the average daily level of carbon monoxide in the air will be $c(p) = 0.5p + 1$ parts per million when the population is p thousand. It is estimated that t years from now the population of the community will be $p(t) = 10 + 0.1t^2$.

1. Express the level of carbon monoxide in the air as a function of time.
2. When will the carbon monoxide level reach 6.8 parts per million.

Composite Functions

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1. Express the level of carbon monoxide in the air as a function of time. $\implies$ find $c(p(t))$.

$$c(p(t)) = 0.5(10 + 0.1t^2) + 1 = 5 + .05t^2 + 1 = 6 + .05t^2$$

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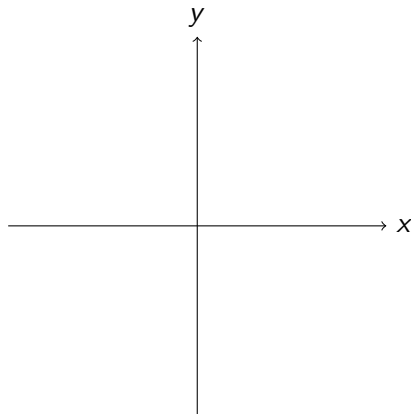
2. When will the carbon monoxide level reach 6.8 parts per million.

$$c(p(t)) = 6.8 = 6 + .05t^2 \implies .8 = .05t^2 \implies t = \left(\frac{.8}{.05}\right)^{\frac{1}{2}} = 4$$

Graphing

The graph of a function f is the spatial or geometric representation of that function. A figure ties together x and $f(x)$ for each x in the domain of f .

A rectangular coordinate system

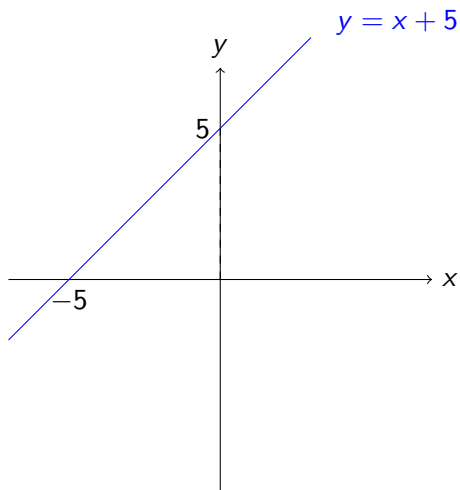


Graphing

For most functions, we can get a good idea of what they look like by plotting points and charting them on a graph. For example, we can graph the function $f(x) = x + 5$ by picking some representative points from the domain, solving their values and plotting them on the coordinate system.

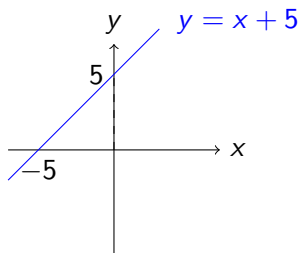
Graphing

x	-3	-2	-1	0
$y = f(x)$	2	3	4	5



Graphing

- ▶ The points where a graph crosses the x -axis are called the **x -intercepts** and where the graph crosses the y -axis the **y -intercepts**.
- ▶ To find the x -intercepts of a function set y equal to zero and solve for x . To find the y -intercept of a function set x equal to zero and solve for y .



Graphing

Find the x and y intercepts

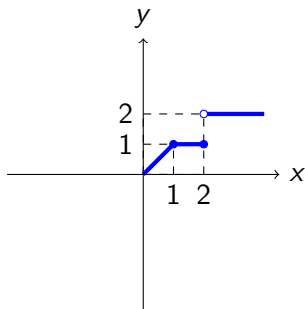
1. $y=x+10$

2. $y=x$

Graphing

We can also plot *stepwise* functions.

$$y = \begin{cases} x & \text{when } 0 \leq x \leq 1 \\ 1 & \text{when } 1 < x \leq 2 \\ 2 & \text{when } 2 < x \end{cases} \quad (7)$$



The open dot indicates that the point is not actually on the graph.

Graphing Non-linear Functions

- ▶ The graph of $Ax^2 + Bx + C$ is a parabola as long as $A \neq 0$. Parabolas given by the above equation have a “U” (or upside down U) shape.

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Graphing Non-linear Functions

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- ▶ A parabola opens up if $A > 0$ and opens down if $A < 0$.
- ▶ The peak or valley of the parabola is called its **vertex**, and it occurs where $x = \frac{-B}{2A}$.

Graphing Non-linear Functions

- ▶ To graph a parabola, we need three key pieces of information:
 1. The location of the vertex
 2. Whether the parabola opens up or down
 3. Any intercepts

Graphing Non-linear Functions

Example: Let's graph $y = x^2 + 3$.

1. Identify A , B , and C . $A = 1$, $B = 0$, $C = 3$.
2. The location of the vertex: $x = \frac{-B}{2A} = \frac{0}{2} = 0$. Now plug zero into get the y coordinate. $y = 0^2 + 3 = 3$ The vertex is at $(0,3)$.
3. Whether the parabola opens up or down: up since $A > 0$
4. Any intercepts:

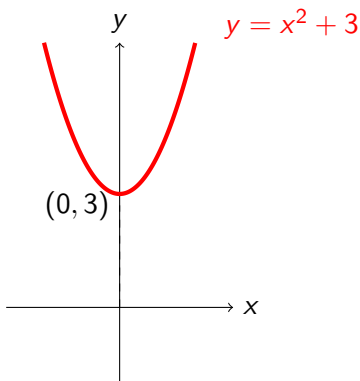
$$x = \frac{0 \pm \sqrt{0 - 4(1)(3)}}{2} = x = \frac{\pm \sqrt{-12}}{2} \implies \text{complex solution.}$$

$$\text{Or solve directly from } x^2 + 3 = 0 \implies x = \pm \sqrt{-3}.$$

No real roots $\implies$ does not cross x -axis

Graphing Non-Linear Functions

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Graphing Non-Linear Functions

Example: Suppose the relationship between age (x) and earnings (y) can be expressed as $y = 50x - \frac{1}{2}x^2$.

1. At what age do earnings peak?
2. Graph the function and find intercepts

Suppose the relationship between age (x) and earnings (y) can be expressed as $y = 50x - \frac{1}{2}x^2$.

1. At what age do earnings peak?

Identify $A, B, C \rightarrow A = -\frac{1}{2}, B = 50, C = 0$.

Find the vertex $x = \frac{-50}{-1} = 50$. So, earnings peak at age 50.

To find y coordinate plug 50 into the function

$$y = 50(50) - \frac{1}{2}(50)^2 = 1250 \implies \text{peak } (x, y) = (50, 1250)$$

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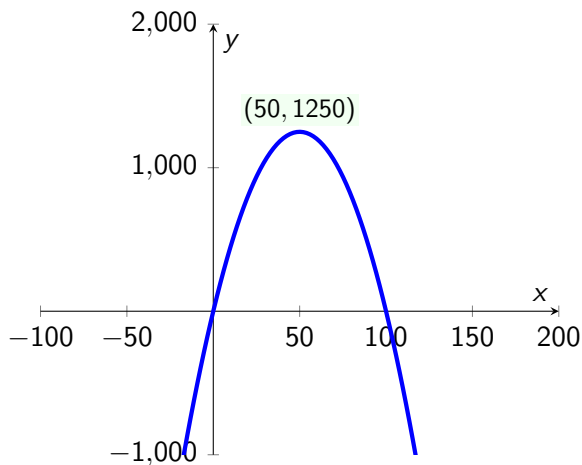
2. Find intercepts

$$x = \frac{-50 \pm \sqrt{50^2 - 4(-.5)(0)}}{-1} = x = \frac{-50 \pm \sqrt{2500}}{-1} = \frac{-50 \pm 50}{-1}.$$

$$x = \frac{-50+50}{-1} = 0 \text{ and } x = \frac{-50-50}{-1} = 100$$

The function crosses the x -axis at 0 and 100. The peak is at (50, 1250) and we know it opens down because $A < 0$

Graphing Non-Linear Function



Intersection of Functions

Intersection of Functions

The intersection of functions on a graph gives us key information. Specifically, the values of x for which two functions $f(x)$ and $g(x)$ intersect.

To find this algebraically, set $f(x)$ equal to $g(x)$ and solve for x .

Example: Find all intersection points for $f(x) = 5x + 3$ and $g(x) = 3x$

Example: Intersection Points

$$\begin{aligned}\text{Set } f(x) = g(x) &\implies 5x + 3 = 3x \\ &\implies 2x = -3 \\ &\implies x = \frac{-3}{2}\end{aligned}$$

To find y , plug into either function:

$$g\left(\frac{-3}{2}\right) = \frac{-9}{2}$$

Application to Demand

Application to Demand

We can use graphing and finding intersections for Supply and Demand models in microeconomics.

Quantity demanded is the amount of goods buyers are willing and able to purchase.

$$\textbf{Law of Demand: } P \uparrow \implies q^D \downarrow$$

Demand Schedule for Smoothies

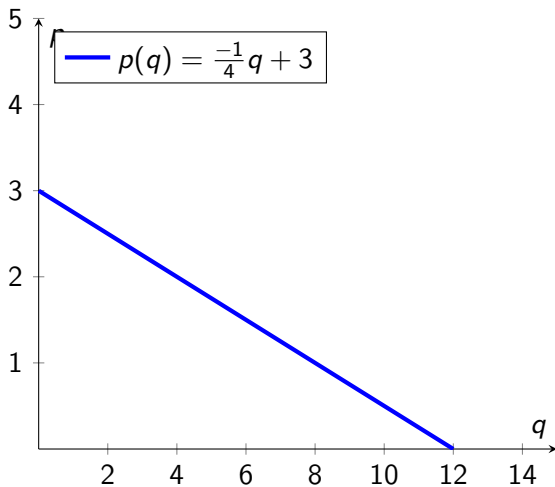
price (\$)	quantity
0	12
0.50	10
1	8
1.50	6
2	4
2.50	2
3	0

Demand Function

The function representing these values can be expressed as:

$$p(q) = \frac{-1}{4}q + 3$$

Demand Curve



Market Demand

price (\$)	Catherine (q_1)	Nicholas (q_2)	Market
0	12	7	19
0.50	10	6	16
1	8	5	13
1.50	6	4	10
2	4	3	7
2.50	2	2	4
3	0	1	1

Market Demand Function

To analyze markets, we need the market demand, which is the sum of all individual demands:

$$Q_{mkt} = \sum_{\forall i} q_i = q_1 + q_2$$

For Catherine:

$$p(q_1) = -\frac{1}{4}q_1 + 3 \implies q_1 = -4p + 12$$

For Nicholas:

$$p(q_2) = \frac{-1}{2}q_2 + \frac{7}{2} \implies q_2 = -2p + 7$$

Market Demand and Supply Intersection

$$\sum_{\forall i} q_i = -4p + 12 - 2p + 7$$
$$Q_{mkt} = -6p + 19$$

Market Demand and Supply Intersection

Now suppose market supply: $Q_S = 6p - 5$. To find the intersection, set $Q_S = Q_D$:

$$6p - 5 = -6p + 19$$

$$12p = 24$$

$$p^* = 2, \quad Q^* = 7$$

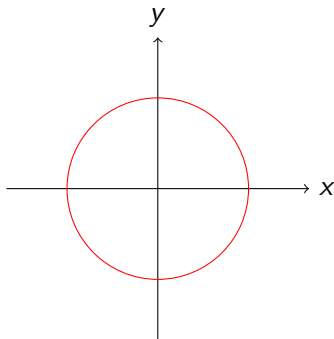
Vertical Line Test

Vertical Line Test

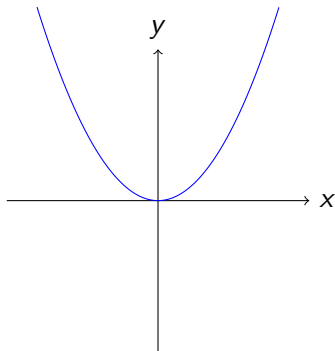
- ▶ The vertical line test determines whether or not a curve is a graph of a function.
- ▶ Specifically, a curve is the graph of a function if and only if no vertical line intersects the curve more than once.
- ▶ If an x coordinate has more than one y coordinate, then this function fails the vertical line test.

Vertical Line Test Examples

(a) Fails Vertical Line Test



(b) Passes Vertical Line Test



Horizontal Line Test

- ▶ If a graph fails the vertical line test, it is not a function because it does not satisfy the requirement that each x value maps to exactly one y value.
- ▶ If the horizontal line test fails, then its inverse will fail.

Linear Functions and Slopes

Linear Functions

- ▶ **Definition:** A linear function is of the form: $f(x) = a_0 + a_1x$, where a_0, a_1 are constants.
- ▶ **Slope-Intercept Form:** $y = mx + b$, where m is the slope and b is the y-intercept.
- ▶ Linear functions are traditionally written in the form $y = mx + b$, where $m, b = \text{constants}$ (**slope-intercept form**).

Slope of a Line

The **slope of the line** measures the “steepness” of the line. The formula for computing the slope is

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

The sign and magnitude of the slope of a line indicate the line's direction and steepness, respectively.

Properties:

- ▶ Positive slope: line rises as x increases.
- ▶ Larger absolute value: steeper line.
- ▶ Smaller absolute value: more gradual slope.

Example: Slope and Y-Intercept

Example: Find the slope and y-intercept of the line $y + 4x = 6$

$$y = -4x + 6 \implies \text{y-intercept is } b = 6 \implies (0, 6)$$

Slope is -4

Point-Slope Form

Another way to represent a line is the **point-slope form of the equation of a line**. Specifically, the equation can be expressed as $y - y_0 = m(x - x_0)$

Example: Find the equation of the line through (6,2) with slope $\frac{1}{5}$.

$$y - 2 = \frac{1}{5}(x - 6) \implies y = \frac{1}{5}x - \frac{6}{5} + 2 \implies y = \frac{1}{5}x + \frac{4}{5}$$

Elasticity and Slope

Continuing with the price and quantity example, we had $P \uparrow$ and $Q_D \downarrow$, but by how much and what can determine how much?

Elasticity and Slope

- ▶ Continuing with the price and quantity example, we had $P \uparrow$ and $Q_D \downarrow$, but by how much and what can determine how much?
- ▶ **Price Elasticity of Demand:** Measures how much quantity demanded responds to price changes.

$$\epsilon_P^D = \frac{\% \Delta Q}{\% \Delta P}$$

e.g., For example suppose that a 10% increase in smoothies $\implies$ quantity $\downarrow$ by 20%. The change in quantity demanded of smoothies is twice as large as the change in price, which is pretty elastic.

Elasticity and Slope

Price Elasticity of Demand: Measures how much quantity demanded responds to price changes.

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Determining Factors:

- ▶ Availability of close substitutes.

Large change in quantity → elastic.

Small change in quantity → inelastic.

Elasticity

- ▶ Necessities vs. Luxury

Elasticity

- ▶ Necessities vs. Luxury
- ▶ Definition of Market

Elasticity

- ▶ Necessities vs. Luxury
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- ▶ Time Horizon

Example: Elasticity Calculation

Example

Points:

$$B : P = \$6, Q = 80$$

$$A : P = \$2, Q = 120$$

Compute ϵ_p :

Example: Elasticity Calculation

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$$B : P = \$6, Q = 80$$

$$A : P = \$2, Q = 120$$

Compute ϵ_p :

$$1. B \rightarrow A: \epsilon_p = \frac{120-80}{80} / \frac{2-6}{6} = \frac{1/2}{-(2/3)} = -\frac{3}{4}$$

Example: Elasticity Calculation

Example

Points:

$$B : P = \$6, Q = 80$$

$$A : P = \$2, Q = 120$$

Compute ϵ_p :

$$1. A \rightarrow B: \epsilon_p = \frac{80-120}{120} / \frac{6-2}{2} = \frac{-(1/3)}{2} = -\frac{1}{6}$$

Mid-Point Method for Elasticity

To address base calculation issues:

$$\bar{Q} = \frac{Q_\theta + Q_\gamma}{2} = 100 \quad \text{and} \quad \bar{P} = \frac{P_\theta + P_\gamma}{2} = 4$$

$$\therefore \epsilon_p = \frac{120 - 80}{\bar{Q}} / \frac{2 - 6}{\bar{P}} = \frac{\frac{40}{100}}{(-1)} = -\frac{40}{100} = -\frac{2}{5}$$

Arc Elasticity: $\frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}$

Elasticity and Demand Curve

Demand Elasticity:

- ▶ $|\epsilon_P^D| > 1 \Leftrightarrow$ Elastic $\Leftrightarrow$ Flatter demand curve.
- ▶ $|\epsilon_P^D| < 1 \Leftrightarrow$ Inelastic $\Leftrightarrow$ Steeper demand curve.
- ▶ $|\epsilon_P^D| = 1 \Leftrightarrow$ We consider the elasticity unit elastic, if the percent change in quantity = percent change in price (ie. 1 to 1 type change).

Note: Report ϵ_P^D as a positive number for clarity.

e.g., If $\epsilon_P^A = -2$ and $\epsilon_P^B = -1/2$. If I say the larger number is more

elastic. Technically ϵ_P^B , but we care about magnitude. Therefore we just report absolute value to be on same page

Elasticity Changes Along Demand Curve

As you move down the demand curve, ϵ_p becomes more inelastic.

Intuition:

- ▶ High prices: slight price increase $\implies$ stop buying.
- ▶ Lower prices: same increase is not as significant.

A 10% increase at \$2 is \$2.2 $\implies$ 20 cents. A 10% increase at \$1 million is \$1.1 million $\implies$ \$100,000

Slope and Elasticity

$$\epsilon_P^D = \frac{\% \Delta Q}{\% \Delta P} \implies \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Recall:

$$\text{Slope} = \frac{\Delta Y}{\Delta X} \implies \frac{\Delta P}{\Delta Q} \implies \text{Inverse of } \frac{\Delta Q}{\Delta P}$$

$\implies$ Slope $\neq \epsilon_P^D$ since $\frac{\Delta Q}{\Delta P}$ is constant, but ϵ_P^D changes along the demand curve, depending on $\frac{P}{Q}$.

Parallel and Perpendicular Lines

Specifically, let m_1 and m_2 be the slopes of the nonvertical lines L_1 and L_2 . Then,

- ▶ L_1 and L_2 are **parallel** iff $m_1 = m_2$ L_1 and L_2 are **perpendicular** iff $m_2 = -\frac{1}{m_1}$

As an example let L be the line $x + y = 2$. Find the equation of a line parallel to L through point $(1,2)$. The slope of L is -1 . Use point-slope formula.

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$$y - 2 = (-1)(x - 1) \implies y = -x + 1 + 2 \implies y = -x + 3$$

Exponential Functions

Exponential Functions

- ▶ In previous lessons, we dealt with power functions such as $y = x^2$, in which the variable x is raised to a constant power 2 (the exponent).
- ▶ Now we'll be looking at functions for which a constant base like 2 is raised to a variable x so $y = 2^x$. These are called **exponential functions**.

Exponential Functions

Definition: An exponential function with base b is defined by

$$f(x) = b^x \quad \text{for } b > 0$$

Properties of Exponential Functions

For bases $a, b > 0$ and any real numbers x, y :

- ▶ Equality Rule: $b^x = b^y$ iff $x = y$
- ▶ Product Rule: $b^x b^y = b^{x+y}$
- ▶ Quotient Rule: $\frac{b^x}{b^y} = b^{x-y}$
- ▶ Power Rule: $(b^x)^y = b^{xy}$
- ▶ Multiplication Rule: $(ab)^x = a^x b^x$
- ▶ Division Rule: $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$

Examples

Simplify the following:

1. $s^2 s^{-1}$

2. $\frac{b^{2x}}{b^{4x}}$

3. $(8^{\frac{1}{2}})(6^{\frac{1}{2}})$

Examples

Simplify the following:

1. $s^2 s^{-1} = s$

2. $\frac{b^{2x}}{b^{4x}} = b^{2x-4x} = b^{-2x}$

3. $(8^{\frac{1}{2}})(6^{\frac{1}{2}}) = (8 \cdot 6)^{\frac{1}{2}} = 48^{\frac{1}{2}}$

Natural Exponential Base

Exponential base e : In calculus, the most common exponential base is denoted by the letter e .

- ▶ e is an irrational number approximately equal to 2.72.
- ▶ Frequently referred to as the "natural exponential base."

Logarithmic Functions

- ▶ If N is a positive number and b is also positive (with $b \neq 1$), we may ask if there is a number L such that $b^L = N$. It can be shown that there is exactly one such number L .

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- ▶ L is the exponent that b must be raised to yield N . If instead of constant N , we can let the number of N vary (call it x instead of N).
- ▶ The **logarithm** of x to the base b is y such that $b^y = x$ and we write $y = \log_b x$ (where $b > 0$ and $b \neq 1$). In the special case where $b = e$, the number y is such that $e^y = x$ and is called the natural logarithm $y = \ln x$.

Logarithmic Functions

Definition:

- ▶ $L = \log_b N$ iff $b^L = N$
- ▶ $y = \log_b x$ iff $b^y = x$
- ▶ For $b = e$, $y = \ln x$ iff $e^y = x$

Examples

Convert to Logarithms:

1. $81 = 9^2$

2. $\frac{1}{9} = 3^{-2}$

Solve for the unknown quantity:

3. $y = \log_{30} 900$

4. $y = \log_2 \frac{1}{32}$

Examples (continued)

Solve for the unknown quantity:

1. $81 = 9^2 \implies \log_9 81 = 2$

2. $\frac{1}{9} = 3^{-2} \implies \log_3 \frac{1}{9} = -2$

3. $y = \log_{30} 900$

$$30^y = 900 \implies y = 2$$

4. $y = \log_2 \frac{1}{32}$

$$2^y = \frac{1}{32} \implies 2^{-5} = \frac{1}{32} \implies y = -5$$

How many times do you have to raise 2 to in order to get 32 in the denominator? $5 \implies y = -5$

Neutralizing Effect

- ▶ The natural logarithm ($\ln x$) and the expression e^x have a “neutralizing” effect (cancel each other out; note that $e^{\ln x} = x$ requires $x > 0$).

$$\ln e^x = e^{\ln x} = x$$

- ▶ The natural log of e raised to x is simply x and e raised to the power $\ln x$ is simply x .
- ▶ This simply tells us that the consecutive impact on x of the exponential function followed by the natural log function leaves x unchanged.

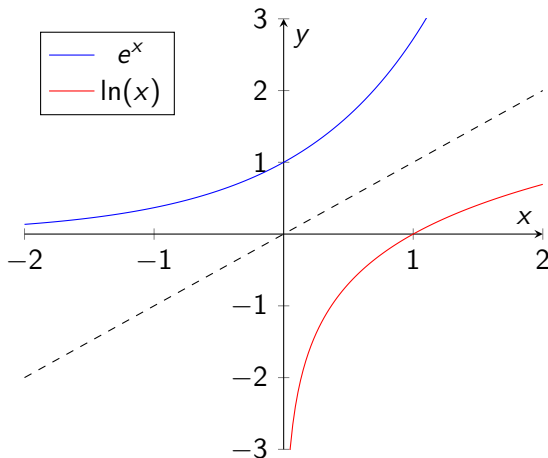
Neutralizing Effect

- ▶ Claim is that $\ln e^x = x$.
- ▶ Let $\log_e e^x = u$. Recall that in this case $b = e$, $L = u$, and $N = e^x$.
- ▶ Remember that we interpret L (or u) as what b or (e) needs to be raised to in order to get N (or e^x).
- ▶ In other words, $b^L = N$ or in this case $e^u = e^x$. Now, recall the equality rule $s^x = s^y$ iff $x = y$.
- ▶ This means $u = x \implies \log_e e^x = x \implies \ln e^x = x$

Inverse Functions

Inverse functions: f and g are inverses if $f(g(x)) = x$ and $g(f(x)) = x$.

Exponential and logarithmic functions are inverses.



Properties of Logarithms

For all logarithms:

- ▶ Equality Rule: $\log u = \log v$ iff $u = v$
- ▶ Product Rule: $\log(uv) = \log u + \log v$
- ▶ Power Rule: $\log u^r = r \cdot \log u$
- ▶ Quotient Rule: $\log \frac{u}{v} = \log u - \log v$

Proof of Product Rule

Let's prove the product rule:

Let $m = \log_b u$ and $n = \log_b v \implies u = b^m$ and $v = b^n$

$$\log_b(uv) = \log_b(b^m b^n) = \log_b(b^{m+n}) = m + n = \log_b u + \log_b v$$

Examples

Solve for x in terms of y :

1. $\log_a x = y^3$

$$a^{\log_a x} = a^{y^3} \implies x = a^{y^3}$$

2. $\log_a x = \log_a 3 + \log_a y$

$$\log_a x = \log_a(3y) \implies x = 3y$$

Examples

Solve for x in terms of y :

1. $\ln x = \log_a y$

2. $y = ge^{hx}$

Simplify

3. $e^{3 \ln x}$

4. $e^{\frac{1}{2} \ln 6x}$

Use log properties to write as sums, differences, or products

5. $\ln 76x^3$

6. $\ln x^5 y^2$

7. $\ln 12 + \ln 5 - \ln 6$

8. $5 \ln \frac{1}{2}$

Examples

Solve for x in terms of y :

1. $\ln x = \log_a y \implies e^{\ln x} = e^{\log_a y} \implies x = e^{\log_a y}$

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Solve for x in terms of y :

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2. $y = ge^{hx} \implies \ln y = \ln(ge^{hx}) \implies \ln y = \ln g + \ln(e^{hx}) \implies$
 $\ln y - \ln g = hx \implies x = \frac{\ln y - \ln g}{h}$

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3. $e^{3 \ln x} = e^{\ln x^3} = x^3$

4. $e^{\frac{1}{2} \ln 6x} = e^{\ln(6x)^{\frac{1}{2}}} = (6x)^{\frac{1}{2}}$

Use log properties to write as sums, differences, or products

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$$5. \ln 76x^3 = \ln(76) + \ln x^3 = \ln(76) + 3 \ln x$$

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Examples

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Simplify

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$$8. 5 \ln \frac{1}{2} = 5(\ln 1 - \ln 2)$$